



Capstone Project - Final Report

AnaChart

**Exploiting Analyst Recommendation Patterns
with Bayesian Skill Modeling to Generate
Returns**

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1. Introduction

Equity analysts occupy a critical position in modern financial markets, serving as intermediaries between complex corporate information and actionable investment decisions. Through their stock recommendations, rating revisions, and price target updates, analysts collectively influence billions of dollars in capital allocation across institutional and retail portfolios. Yet despite their market influence, the predictive value of individual analyst recommendations remains heterogeneous, some analysts consistently move markets while others contribute little beyond noise.

This project leverages the AnaChart dataset, a comprehensive collection of analyst ratings, price targets, and recommendation revisions covering Nasdaq-100 constituents, to systematically evaluate analyst signal quality and construct a Bayesian-driven investment strategy. The work is grounded in the theoretical framework of Jaisson et al. (2021) (1), who demonstrate that while the aggregate informativeness of analyst recommendations has remained stable over time, the speed at which markets incorporate this information has accelerated dramatically. This has rendered naive strategies of simply buying upgrades and selling downgrades increasingly unprofitable, making the identification of underestimated analysts a more valuable and nuanced objective.

The project proceeds in two interconnected stages. The first stage develops a comprehensive exploratory dashboard that synthesizes analyst activity, recommendation dynamics, price target behavior, and broker-level patterns across time, sector, and coverage universe. This provides the empirical foundation necessary to understand the structural properties of the dataset before modeling.

The second stage implements the core investment methodology: a Bayesian ridge regression model to estimate individual analyst strength scores based on post-recommendation price drift. Critically, the strategy distinguishes between two types of analyst influence. High-impact analysts whose recommendations are immediately priced in by the market are identified but deliberately excluded from the trading signal. Instead, the focus is on underestimated analysts: those whose insights are not fully incorporated at the time of recommendation, creating a measurable and exploitable drift in subsequent returns. A long-short strategy is then constructed by going long on upgrades and short on downgrades issued by this select group of underestimated analysts.

The pipeline supporting this analysis spans four stages: raw data retrieval from Google BigQuery, multi-layer data cleaning and feature engineering including rating standardization and recommendation type classification, Bayesian analyst strength scoring and final strategy backtesting against Nasdaq-100 price returns.

By bridging the gap between the theoretical Bayesian graphical modeling framework of Jaisson et al. and an empirically grounded implementation on real analyst data, this project contributes a reproducible, interpretable methodology for analyst signal extraction that is directly applicable to systematic investment decision-making.

2. Literature Review

Barber, Lehavy, and Trueman (2010) (2) examine whether the predictive power of analyst recommendations stems from rating levels, rating changes, or both. Comparing three hedge strategies- levels-only, changes-only, and a combined approach conditioning on both simultaneously, they find that each dimension carries independent predictive value. The combined strategy substantially outperforms either standalone approach, with a dollar invested in 1986 growing to over \$24 by 2006 versus just over \$7 for the individual strategies. Crucially, these results survive momentum controls, contradicting prior literature that suggested level-based returns were not true. The authors also find that both levels and changes predict future unexpected earnings, pointing to genuine information production rather than demand-shifting.

Vukovic, Ugolnikov, and Maiti (2020) (3) investigate whether analyst recommendations generate abnormal returns on the NYSE and LSE using event study methodology over a five-day window. Employing a one-factor market model with the S&P 100 and FTSE 100 as benchmarks, they find that positive recommendations produce significant abnormal returns on and immediately after the publication date, while negative recommendations trigger a shorter-lived but still significant negative reaction. Neutral recommendations generate no meaningful price response. Notably, both markets behave nearly identically. The authors also find that upgrades produce a sharper but less durable market reaction than reiterations of existing positive ratings, and that larger firms attract stronger responses to neutral and negative recommendations, likely reflecting greater brokerage attention. Their conclusion that developed markets are not strongly efficient underpins the assumption here that analyst signals remain exploitable.

Lee, Hsieh, and Miao (2024) (4) evaluate the accuracy of analyst target prices on the Taiwan Stock Exchange across five dimensions: directional correctness, target achievement, signed error, absolute error, and overshooting; comparing analyst forecasts against four alternative models ranging from naive historical projection to a market model with perfect foresight. Despite a systematic upward bias of 9.4% and a large absolute pricing error of 24.8%, analyst forecasts consistently outperform all historically-based models and perform comparably to the perfect-foresight benchmark, suggesting that analysts contribute genuine informational value beyond what past data alone can capture. However, accuracy decays rapidly after issuance, supporting the use of analyst targets as short-term rather than twelve-month guides. At the cross-sectional level, accuracy improves with brokerage experience and peer competition, while aggressive forecasts and high stock volatility are associated with larger errors.

Jaisson, Nguyen, Messikh, and Susinno (2021) (1) investigate the predictive power of sell-side analyst recommendations and their impact on stock prices using the I/B/E/S database from Refinitiv, decomposing post-recommendation price behavior into two distinct components: an Impact window spanning the day before to the day after the recommendation, and a Drift window covering day +1 to day +20. Applying this decomposition across three eight-year periods from 1994 to 2020, they document a structural shift whereby the Impact component has strengthened over time while the Drift component has steadily shrunk, attributing this to arbitrage dynamics that have compressed post-recommendation return predictability into the immediate announcement window. Their central methodological contribution is a Bayesian ridge

regression model that estimates a latent strength parameter for each analyst, regularizing estimates toward zero for analysts with limited recommendation histories to avoid overfitting on small samples. The framework distinguishes between impactful analysts whose recommendations are immediately priced in by the market, leaving no exploitable residual and underestimated analysts, whose recommendations generate persistent post-announcement drift approximately twice that of weaker analysts out of sample. The authors conclude that naive upgrade-downgrade strategies are barely profitable after costs in recent years and that selectively targeting underestimated analysts represents the more viable systematic approach. This Bayesian strength estimation framework and the impact-drift decomposition directly underpin the methodology employed in the present project.

Karminsky, Polozov, and Suffert (2024) (5) investigate whether analyst recommendations generate abnormal returns on the Nasdaq-100 and whether star-ranked analysts produce more profitable signals than their non-star counterparts, employing long, short, and long-short portfolio strategies constructed from buy and sell recommendations evaluated against CAPM, Fama-French three-factor, and Carhart four-factor benchmarks. Their results confirm that all analyst groups stars and non-stars alike generate statistically significant abnormal returns, challenging the Efficient Market Hypothesis and affirming that analyst signals retain exploitable informational content. However, the performance differential between star and non-star analysts is asymmetric across portfolio types: in short portfolios, star analysts outperform non-stars by a monthly alpha differential of 0.55%, while in long portfolios no statistically significant difference is found, attributed to positivity bias and the heavy institutional attention already directed at large-cap Nasdaq-100 names. Analysts ranked first by StarMine demonstrate the most consistent outperformance across all portfolio categories. The study further finds that investor sentiment, proxied by the Nasdaq Volatility Index within a GARCH(1,1) framework, exerts a statistically significant positive effect on return volatility, with bidirectional causation between sentiment and volatility observable over a five to ten day window. These findings are directly relevant to the present project in two respects: they confirm the persistence of analyst signal value in the Nasdaq-100 universe specifically, and they motivate the selective targeting of high-quality analysts analogous to the underestimated analyst identification approach of Jaisson et al. (2021) (1) rather than treating all recommendations as equally informative.

Chen, Xu, and Zhang (2023) (6) investigate the predictability of analyst rating quality (ARQ), defined as a binary outcome capturing whether a recommendation correctly predicted the rated stock's movement within the analyst's own stated time horizon and benchmark, using approximately 124,000 analyst ratings from the Chinese CSMAR database spanning 2015 to 2019. Constructing 47 candidate predictors across analyst, rating, and stock characteristics, their primary methodological contribution is ComEFS, a hierarchical ensemble feature selection framework that clusters 18 base selectors into methodologically similar communities using the Louvain algorithm before aggregating within and across communities sequentially, preventing correlated selection methods from dominating the final feature ranking. ComEFS outperforms all 24 benchmark methods across standard classification metrics regardless of downstream classifier, with top predictors dominated by brokerage size and report volume, relative rating positioning versus peer consensus, earnings per share, and return volatility. Partial dependence analysis reveals that analysts with narrower coverage produce higher quality ratings, that contrarian recommendations carry greater informational content, and that stock volatility amplifies the informational advantage of large brokers. The feature taxonomy and ARQ measurement methodology provide a validated blueprint for predictor construction in the

present project, particularly the brokerage-level and rating-level variables directly available in the AnaChart Nasdaq-100 dataset.

Zaremba and Konieczka (2015) (7) provide out-of-sample evidence on the profitability of analyst recommendations using data from the Warsaw Stock Exchange between 2004 and 2013. Comparing target-price-based and rating-based strategies across equal- and value-weighted portfolios, the authors find that recommendations carry genuine predictive content, but that this value is highly sensitive to how portfolios are constructed. Most notably, profitability is concentrated among small-cap stocks and only materializes as statistically significant abnormal returns (alphas of 0.88%–1.34% across the CAPM, Fama-French, and Carhart models) when portfolios are equally weighted; value-weighted portfolios show no significant abnormal performance. The study also finds that momentum and value factors partially explain the observed returns, suggesting analysts tend to favor stocks that have already performed well or trade at high book-to-market ratios, rather than purely identifying mispriced securities. Importantly, the authors caution that these results exclude transaction costs and liquidity constraints, raising questions about real-world implementability.

3. Research Objectives

This project aims to evaluate the informational value of equity analyst recommendations on Nasdaq-100 constituents using the AnaChart dataset, and to construct a Bayesian-driven investment strategy targeting underestimated analysts whose signals are not fully incorporated by the market at the time of issuance. The study is guided by the following specific objectives:

- **Construct a Robust Analyst Recommendation Pipeline**
Design and implement a four-stage data pipeline covering raw data retrieval from Google BigQuery, multi-layer cleaning and standardization of analyst ratings and price targets, recommendation type classification spanning initiations, upgrades, downgrades, reiterations, and price target revisions, and feature engineering to prepare structured inputs for downstream modeling.
- **Decompose Post-Recommendation Price Behavior into Impact and Drift**
Following the framework of Jaisson et al. (2021), measure the immediate market reaction to analyst recommendations within a two-day impact window and the subsequent post-announcement drift over a one to twenty day horizon. Assess whether a statistically and economically significant drift component remains exploitable in the Nasdaq-100 universe in recent years.
- **Estimate Bayesian Analyst Strength Scores**
Apply a Bayesian ridge regression model to estimate latent strength parameters for individual analysts using rolling five-year windows, regularizing estimates toward zero for analysts with limited recommendation histories. Distinguish between impactful analysts whose recommendations are immediately priced in and underestimated analysts whose signals generate persistent post-announcement drift.
- **Design and Backtest an Underestimated Analyst Investment Strategy**
Construct a long-short portfolio strategy that goes long on upgrades and short on

downgrades issued by analysts identified as underestimated by the Bayesian strength model. Backtest the strategy against passive Nasdaq-100 benchmarks and evaluate performance using risk-adjusted return metrics including Sharpe ratio, maximum drawdown, and volatility

- **Generate Actionable Insights for Analyst Signal Interpretation**

Translate empirical findings into practical guidance on which analyst signals carry genuine informational value, how Bayesian strength scoring can be used to filter recommendations in real time, and how investors can integrate analyst-driven signals into systematic portfolio construction.

4. Data Exploration and Tools

Dataset: AnaChart NASDAQ-100 Stock Analysts Price Targets and Ratings

Other data sources

- *yfinance*: for adjusted close price of stocks and QQQ benchmark
- *FRED*: for T-Bill rates

Tools

- Python: data cleaning and preprocessing pipeline, analysis, strategy tuning, backtesting, visualization
- SQL: data cleaning and preprocessing queries
- Looker Studio: exploratory data analysis and visualization

4.1 Dataset Description

The dataset has 55,106 recommendations across 98 NASDAQ-100 stocks, provided by 1428 analysts. The dataset has recommendations from 23 June 2005 to 12 February 2026.

There are 123 brokers with analyst affiliations. However, 885 analysts are either independent or have an unknown broker affiliation.

4.2 Exploratory Data Analysis

The average price target revision in the data is \$5.34, and has increased by 4.8% from the previous year. The average price target revision percentage is 0.95%, and has increased 11.2% from the previous year.

Recommendation Type Distribution

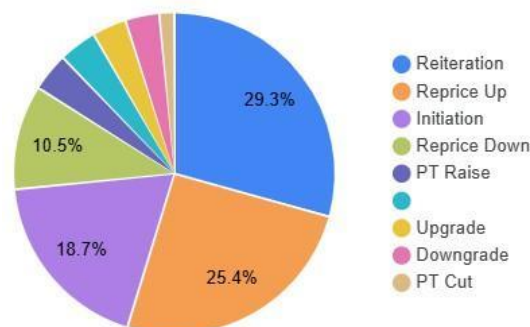


Figure 1: Distribution of Recommendation Types

From Figure 1, we see that 29.3% of the recommendations (16,154 observations) are 'Reiteration'. 'Reprice Up', 'PT Raise', and 'Upgrade' comprise 32.8% of all recommendations (representing 18,038 recommendations), followed by 10,294 initiations. The rest 15.4% of recommendations are either 'Reprice Down', 'Downgrade', or 'PT Cut', which reinforces that analysts tend to have optimism bias.



Figure 2: Distribution of Analyst Ratings

Figure 2 indicates that "Buy" and "Strong Buy" recommendations comprise 72% of all ratings (representing 36,553 observations). This aligns with established literature regarding analyst optimism bias. This bullish tendency is particularly pronounced within the NASDAQ-100, a major index heavily weighted toward large-cap growth stocks.

Rating After / No. of Recommendations					
Rating Before	Buy	Strong Buy	Hold	Strong Sell	Sell
Buy	13,559	73	1,443	10	83
Strong Buy	46	195	3	-	-
Hold	1,519	7	4,693	6	291
Strong Sell	3	-	4	7	1
Sell	45	-	250	1	640
null	20,923	183	5,853	16	925

Figure 3: Analyst Ratings Transition Matrix

From Figure 3, we see that the single largest active transition block is keeping a "Buy" a "Buy" (13,559), and the largest initiation block is "null" to "Buy" (20,923). This shows a massive concentration of total volume in the "Buy" category. It is incredibly rare for an analyst to move a stock directly from a "Buy" to a "Sell" (83 times) or "Strong Sell" (10 times) without pausing at "Hold" first.

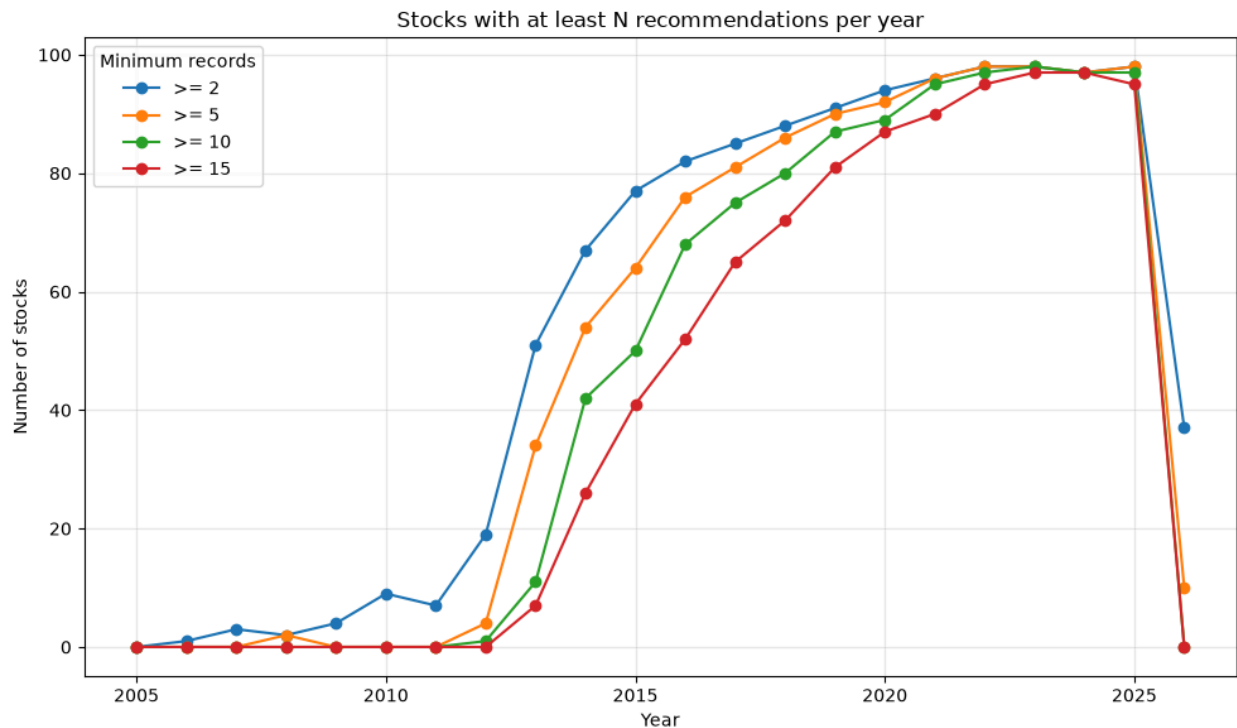


Figure 4: Stock Coverage Over Time

Figure 4 illustrates the growth of stock coverage within the dataset by tracking how many NASDAQ-100 stocks received at least N analyst recommendations in a given year. Coverage was sparse in the early years. Even at the lowest threshold (≥ 2 recommendations), fewer than 20 stocks had meaningful coverage before 2012. Coverage grew sharply across all thresholds, with the ≥ 10 and ≥ 15 recommendation bands converging toward ~95–98 stocks by 2021–2024; effectively full NASDAQ-100 coverage.

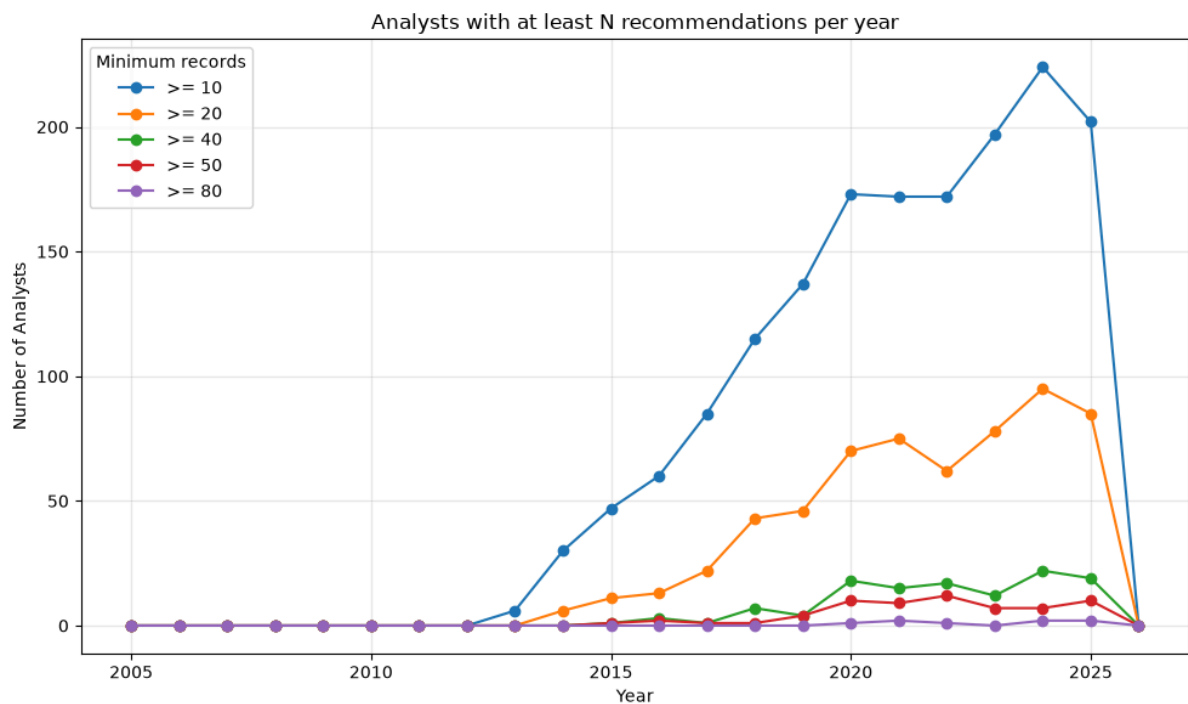


Figure 5: Analyst Activity Over Time

Figure 5 examines how many individual analysts issued at least N recommendations per year, revealing the distribution of analyst productivity. Consistent with the stock-level chart, very few analysts crossed even the ≥ 10 threshold before 2013. Even at peak years (2024–2025), only a small number of analysts reached the ≥ 50 or ≥ 80 recommendation thresholds, while the ≥ 10 group numbered over 200. This confirms that analyst output follows a highly skewed distribution.

The drop in both figures 4 and 5 in the year 2026 is because the dataset covers recommendations till mid-February 2026.

5. Methodology

5.1 Data Cleaning and Preprocessing

The preprocessing workflow in this project was designed to convert various styles of analyst recommendation data into a consistent, analysis-ready format. The pipeline is implemented mainly in BigQuery.

In the first stage, the raw data source table is copied to a writable destination

The second step focuses on text cleaning and type standardization. Some source values contain embedded delimiters or formatting artifacts, so the script keeps only the meaningful part. After that, the data is casted to the appropriate type.

The third step maps many broker-specific recommendation phrases into 5 canonical categories: Strong Buy, Buy, Hold, Sell, and Strong Sell. It also converts these standardized labels into ordinal numeric scores ranging from -2 (Strong Sell) to 2 (Strong Buy), which enables downstream quantitative analysis. To support data quality checks, the script additionally creates an audit table of unmapped rating values so that unusual or previously unseen labels can be reviewed separately.

Finally, a Recommendation Type feature is derived, which classifies events as Initiation, Reiteration, Upgrade, Downgrade, PT Raise, PT Cut, Reprice Up, or Reprice Down.

- Initiations occur when there is no previous rating or price target.
- Upgrade and Downgrade are assigned when both- Rating Before and Rating After are present.
- A PT Raise or PT Cut is assigned if Ratings information is missing or incomplete.
- A Reiteration is assigned if the Rating before and after are the same.
 - However, if the price targets are different, then they are assigned Reprice Up or Reprice Down accordingly.
 - Else, the recommendation type remains Reiteration

5.2 Impact and Drift Decomposition

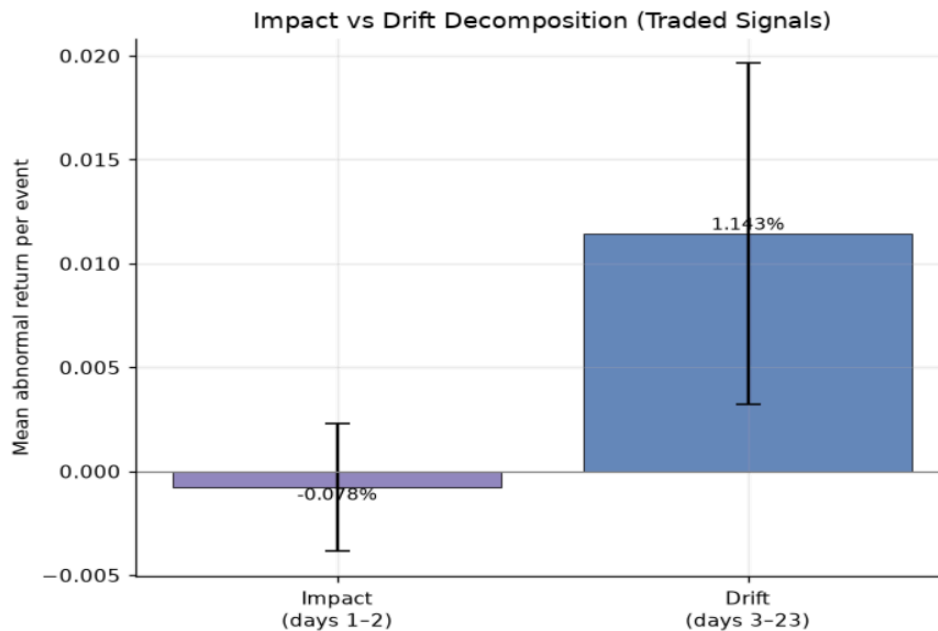


Figure 6: Impact vs Drift Decomposition

Following Jaisson et al. (2021) (1), each recommendation is evaluated along two return windows that separate the immediate market reaction from the delayed, exploitable component of the signal:

- **Impact window** — the return from the day before to the day after the recommendation, measuring how much of the analyst's information is priced in at issuance.
- **Drift window** — the residual price movement in the days that follow.

Why the distinction matters. A recommendation fully reflected in the Impact window leaves no residual to capture and is unprofitable to trade after issuance. One that generates persistent Drift indicates market underreaction: the analyst's signal was not fully incorporated. It is this Drift that the strategy exploits, since it represents a return that can still be captured in the days after the recommendation.

How returns are measured.

- Both returns are directional, aligned to the recommendation sign, so an upgrade contributes positively when the stock rises and a downgrade when it falls.
- Returns are market-adjusted abnormal returns: the stock's log return less the market's over the same interval. This is a beta-one adjustment that removes the common market component while remaining simple.
- Adjusted close prices and each stock's own trading calendar are used.

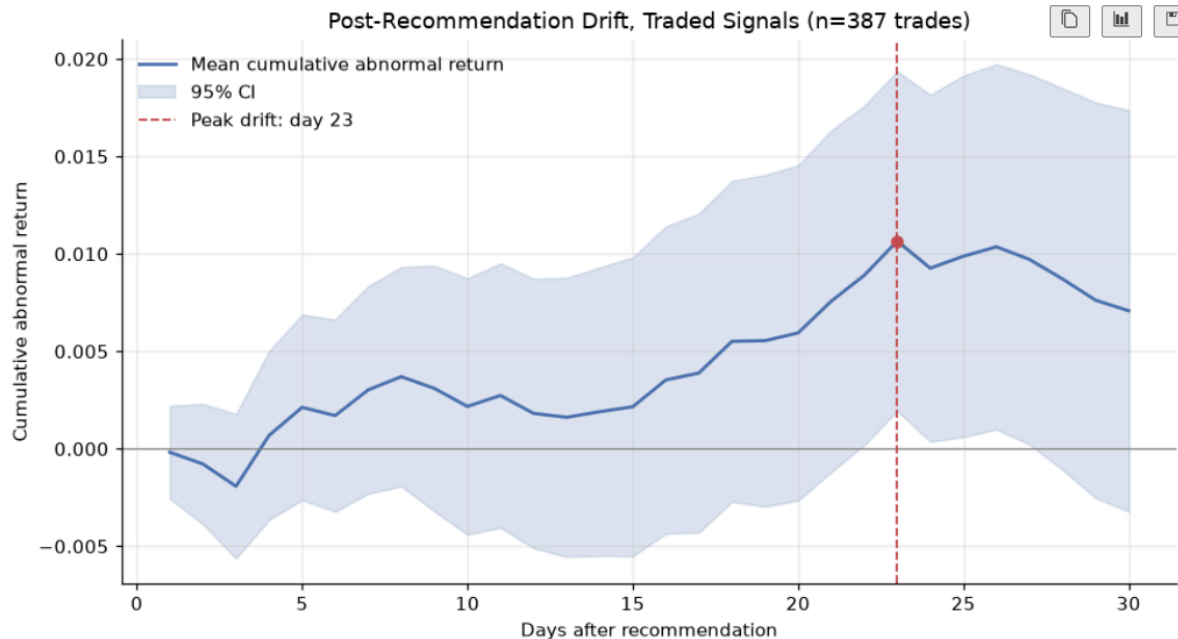


Figure 7: Post Recommendation Drift

Setting the holding horizon empirically. Rather than assuming a holding period, an event study was run across all qualifying recommendations:

- For each recommendation, the market-adjusted abnormal return path was extracted over a fixed horizon from the entry date and sign-aligned to the recommendation direction.
- Averaging cumulative abnormal return across events in event time yields the mean drift profile, with a 95% confidence band from the cross-sectional standard error at each day.
- The day at which cumulative abnormal return peaks and flattens marks the point beyond which no further exploitable drift accrues, and is adopted as the strategy's holding period.

This decomposition reduces each recommendation to a pair of outcome measures, Impact and Drift, which serve as the dependent variables for the Bayesian strength estimation in the next section. It lets the model separate analysts whose value is immediate and arbitrated away from those whose value persists and remains tradeable.

5.3 Bayesian Analyst Strength Model

With each recommendation decomposed into Impact and Drift, the central modeling task is to estimate a latent strength score for every analyst, capturing how much exploitable drift their recommendations generate. The difficulty is that coverage varies enormously: a naive average of realized drift assigns extreme scores to analysts with only a few calls, mistaking noise for skill. The Bayesian ridge formulation of Jaisson et al. (2021) (1) addresses this by shrinking estimates toward the cross-sectional mean in proportion to the evidence available, so an analyst is credited with strength only when their track record supports it.

How strength is computed: Strength is the regularized slope relating an analyst's past recommendation signals to the subsequent sign-aligned drift returns:

- Analysts whose positive signals were consistently followed by positive drift, and negative signals by negative drift, receive high scores; those whose recommendations bore no systematic relationship to returns are pulled toward the mean.
- The numerator accumulates each past signal times its realized drift; the denominator accumulates the squared signals plus a regularization constant encoding a prior of no skill.
- When history is short, the prior dominates and the score stays near the mean. As history grows, the evidence overwhelms the prior and the score converges toward the analyst's true signal-to-drift relationship.

This shrinkage is the key safeguard against overfitting small samples.

Strictly ex-ante estimation: Every score uses only recommendations issued strictly before the date being evaluated, so no information from the recommendation itself or any later one enters its own estimate. This ensures the scores used to weight trades could genuinely have been known at trade time, and that reported performance is not inflated by hindsight. Strength is estimated separately for the Impact and Drift windows, yielding an ex-ante impact strength and an ex-ante drift strength per recommendation.

How the two scores are used. Together they operationalize the project's thesis:

- **High impact, low drift-** the analyst's information is immediately absorbed: informative but not exploitable.
- **High drift-** the analyst's recommendations keep moving the stock after issuance, indicating market underreaction.

It is this second group of underestimated analysts that the strategy targets, using ex-ante drift strength to identify and weight tradeable recommendations.

	Analyst	avg_drift_strength	avg_impact_strength	n_recs	shrunk_drift	shrunk_impact
0	JAMES THALACKER	2.917731	0.384541	22	1.936812	0.324440
1	KEITH BACHMAN	1.531025	0.815008	232	1.460804	0.787149
2	NEAL DINGMANN	2.227249	1.029327	21	1.454704	0.744245
3	TOM O'MALLEY	1.541669	0.390454	22	1.022516	0.328369
4	TIMOTHY HORAN	1.217863	-0.103171	51	0.999031	-0.047964
5	STEPHEN JU	1.015009	1.983956	333	0.982061	1.926529
6	DARREN AFTAH	1.197557	0.114244	34	0.901261	0.136706
7	EVAN SEIGERMAN	1.359461	-0.171512	21	0.887189	-0.041079
8	ARIEL ROSA	1.380388	0.094156	20	0.885461	0.133900
9	MARK KELLEY	1.188073	1.320223	32	0.880475	1.032908
10	GREGORY FRANCFORT	1.435342	0.946265	17	0.865876	0.653449
11	SCOTT GRUBER	1.288151	0.164992	21	0.840553	0.178988
12	GEOFFREY PORGES	0.938997	0.964921	93	0.838207	0.883866
13	ROSS MACMILLAN	0.976807	0.338818	61	0.825464	0.318267
14	DERRICK WOOD	1.002145	0.538750	46	0.806123	0.473904

Figure 8: Top 15 Analysts by Bayesian Strength Model

5.4 Signal Construction and Trade Generation

Each recommendation is encoded into a signed signal reflecting direction and conviction:

- **Upgrade** → **+1**, **Downgrade** → **-1** (rating changes, full conviction)
- **PT Raise** → **+0.5**, **PT Cut** → **-0.5** (price-target revisions without a rating change, half conviction)
- **Initiations**: Buy and Strong **Buy** → **+1**, Sell and Strong **Sell** → **-1**
- **Reiterations and non-directional events** → **0** (no trade generated)

This preserves the distinction between a decisive rating action and a marginal target adjustment, which the position-sizing scheme later exploits.

Each non-zero signal maps to a trade:

- **Direction** = sign of the signal (**+1** long, **-1** short)
- **Entry** = first available close **strictly after** the recommendation date, a look-ahead-safe convention that avoids crediting returns from ratings potentially published after that day's close
- **Exit** = **23 trading days** later (the empirically determined drift horizon), using each stock's own trading calendar so holidays do not distort the holding period

Trades are then expanded into a daily position book:

- Each trade contributes its direction to every day it is held
- Overlapping positions in the same stock from multiple concurrent recommendations are **summed**, so conviction reinforced by several analysts produces a larger net position

5.5 Position Sizing

Once trades are generated, each must be assigned a capital weight. Two schemes are implemented and compared, sharing identical trade direction, entry, and exit so that any performance difference is attributed to sizing alone.

Baseline (equal weight)

- Every trade receives an identical weight of **1.0**, regardless of signal strength
- A decisive rating change and a marginal price-target revision are treated identically
- Serves as the reference point: a sizing scheme earns its complexity only if it beats equal weight

Signal magnitude

- Each trade is weighted by the absolute value of its encoded signal
- Full rating changes (magnitude **1.0**) receive twice the capital of price-target-only revisions (magnitude **0.5**)
- Premise: a rating action reflects stronger conviction than a target tweak and should carry proportionally more risk
- Because the magnitude is known at the moment of the recommendation, this introduces

no look-ahead and requires no external estimation, the most robust step beyond equal weighting.

5.6 Backtesting Framework (2023-2025) and Metrics

The daily position book is valued and evaluated through a backtesting framework that converts positions into a return series and summarizes performance across return, risk, and trade-level dimensions.

Mark-to-market valuation

- Each day, every open position is matched to that stock's realized return for the day
- The portfolio return is the position-weighted average of individual stock returns, divided by gross exposure so the result is a properly scaled daily return
- Positions with missing price data on a given day are excluded from both the weighted return and the exposure denominator, so a single missing price cannot distort the day's return
- Compounding the daily returns produces the strategy's equity curve

Transaction costs (10 bps)

- Costs are applied on **turnover**, the day-over-day change in dollar positions, not on positions merely held
- Each unit of turnover is charged a fixed cost in basis points, so a low-turnover book is penalized less than a high-turnover one
- This ensures reported performance is net, not gross, and that a strategy which churns frequently cannot appear artificially profitable

Performance metrics.

The strategy is evaluated on four fronts:

- *Return*: annualized (geometric) return, and cumulative equity growth
- *Risk-adjusted*: Sharpe ratio (excess return per unit of total volatility) and Sortino ratio (excess return per unit of downside deviation, penalizing only losses)
- *Drawdown*: maximum peak-to-trough decline, and the Calmar ratio (annualized return over maximum drawdown)
- *Trade-level*: hit rate (fraction of individual trades that were profitable) and win-loss ratio (average win size over average loss size)

Return and risk-adjusted metrics are computed on the daily portfolio return series, while trade-level metrics are computed per individual trade, so they measure signal quality independently of how positions are sized or overlapped.

6. Results

Both variants of the strategy generated positive risk-adjusted returns over the backtest period, with signal-magnitude sizing improving on the equal-weight baseline across every return and risk-adjusted metric.

Metric	Baseline (equal weight)	Signal Magnitude	NASDAQ-100 Benchmark
Annualized Return	16.52%	20.60%	27.32%
Annualized Volatility	14.97%	14.90%	19.79%
Sharpe Ratio	0.783	1.017	1.008
Sortino Ratio	1.207	1.634	1.469
Maximum Drawdown	-13.20%	-14.73%	-23.38%
Calmar Ratio	1.251	1.399	1.168
Hit Rate	53.49%	53.49%	—
Win-Loss Ratio	1.020	1.020	—

Figure 9: Backtest Results (2023-2025)

Strategy	Sharpe (point est.)	90% CI Lower	90% CI Upper
Baseline (equal weight)	0.783	0.133	2.028
Signal Magnitude	1.017	0.398	2.258

Figure 10: Bootstrap 90% Confidence Intervals for the Sharpe Ratio

To test whether the risk-adjusted performance could be an artifact of chance, the Sharpe ratio of each strategy was bootstrapped by resampling the daily returns, producing a 90% confidence interval.

- For both strategies the interval lies **entirely above zero** (baseline [0.13, 2.03]; signal magnitude [0.40, 2.26]), so the Sharpe ratio is statistically distinguishable from zero at the 90% level and unlikely to be a chance result.
- Signal magnitude has a **higher lower bound** (0.40 vs 0.13), indicating a more reliably positive edge than the baseline.
- Both intervals are **wide**, reflecting the limited length of the backtest: the sign and significance of the edge are established, but its precise magnitude carries meaningful estimation uncertainty.

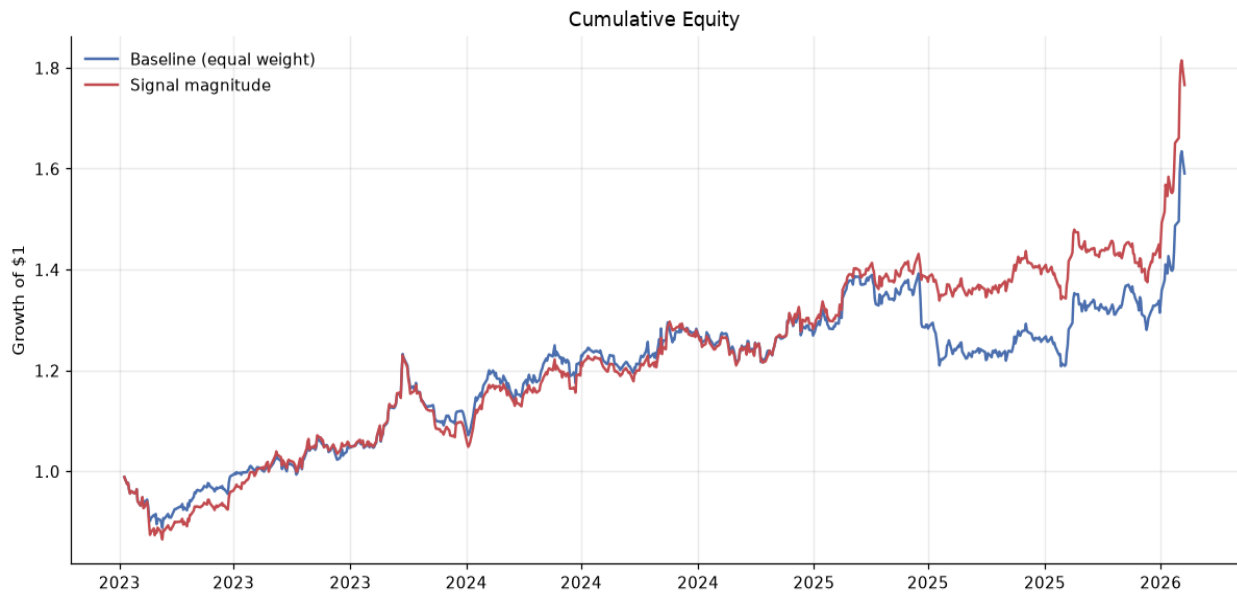


Figure 11: Cumulative Equity Curve (Baseline vs Signal Magnitude)

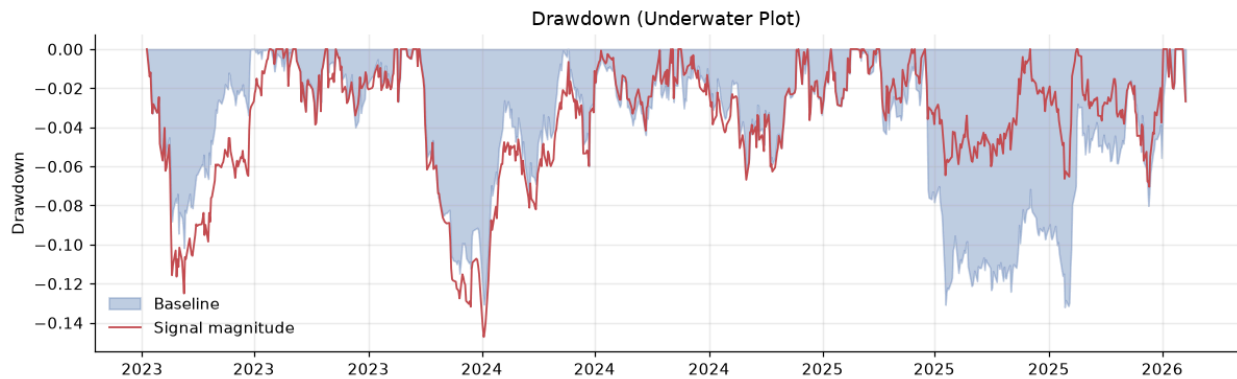


Figure 12: Drawdown Curve (Baseline vs Signal Magnitude)

Both strategies compounded a starting dollar to roughly 1.6–1.8 $\times$ over the three-year backtest, a sustained edge rather than a single lucky episode.

For most of the period the two schemes track closely. They diverge from mid 2025, when the equal-weight baseline (blue) takes a sharp drawdown that signal-magnitude sizing (red) largely avoids, holding a lead through 2025. This episode drives magnitude's higher return (20.6% vs 16.5%) and Sharpe (1.02 vs 0.78): weighting high-conviction rating changes more heavily, and marginal price-target revisions less, protected the book during stress. Both curves rally into early 2026, ending near 1.8 $\times$ (magnitude) and 1.6 $\times$ (baseline).

The gradual divergence, rather than a single jump, suggests magnitude sizing helps structurally by down-weighting weaker signals, not through luck on one trade.

Statistic	Baseline (equal weight)	Signal Magnitude
Annualized Alpha	10.0%	14.1%
Alpha t-statistic	1.161	1.645
Alpha p-value	0.246	0.100
Beta	0.071	0.043
Beta t-statistic	2.621	1.583
Beta p-value	0.009	0.114
Observations (days)	765	765

Figure 13: Alpha-Beta Regression Against NASDAQ-100 Benchmark

To separate genuine analyst-signal alpha from market exposure, both strategies were regressed against a passive NASDAQ-100 (QQQ) benchmark over 765 trading days.

- **Near-zero beta** (0.04 magnitude, 0.07 baseline): the strategy carries almost no market exposure, so returns are effectively independent of the NASDAQ-100, consistent with a market-neutral design.
- **Economically meaningful alpha** (14.1% magnitude vs 10.0% baseline): magnitude sizing improves market-adjusted return, not just raw return.
- **Borderline significance** ($t = 1.65$, $p = 0.10$ for magnitude): the roughly three-year window is short for estimating alpha precisely, widening the standard errors despite sizable point estimates.

Together with the bootstrap Sharpe interval, this supports a genuine but modest edge with very low market dependence, whose exact magnitude remains subject to estimation uncertainty.

Metric (net of costs)	Baseline (equal weight)	Signal Magnitude
Annualized Return	12.65%	16.60%
Annualized Volatility	14.96%	14.90%
Sharpe Ratio	0.558	0.791
Sortino Ratio	0.850	1.255
Maximum Drawdown	-14.50%	-15.20%
Calmar Ratio	0.873	1.092
Hit Rate	53.49%	53.49%
Win-Loss Ratio	1.020	1.020

Figure 14: Backtest Results after Transaction Costs

7. Limitations and Future Work

While the strategy demonstrates a genuine risk-adjusted edge, several limitations qualify the results and point to natural extensions.

Limitations

- **Favorable market regime.** The backtest period was predominantly a bull market for large-cap technology. Although the strategy is market-neutral by construction (beta ~ 0.04), its behavior in a sustained downturn or high-volatility regime is untested.
- **Beta-one abnormal returns.** Impact and drift are measured as market-relative returns (stock return minus market return), which implicitly assumes every stock has a beta of one. For high-beta NASDAQ-100 names this under-adjusts, so a portion of measured drift may reflect residual beta rather than pure analyst-driven movement.
- **Drift concentrated in low-coverage analysts.** The highest drift-strength analysts tend to have shorter recommendation histories, precisely the group the Bayesian shrinkage is designed to distrust. The ≥ 10 -recommendation filter and shrinkage mitigate this, but the exploitable signal rests partly on analysts with thinner track records.
- **Transaction cost simplification.** Costs are modeled as a flat basis-point charge on turnover and do not capture market impact, bid-ask spread variation across names, or borrow costs on the short leg, which would matter for a live long-short implementation.
- **Universe scope.** The strategy is confined to NASDAQ-100 constituents, a heavily-covered, large-cap universe where analyst signals are most rapidly arbitrated.

Future Work

- **Full market-model abnormal returns.** Replace the beta-one adjustment with per-stock betas estimated over a pre-event window, isolating analyst-driven drift from residual market exposure.
- **Extended and out-of-sample testing.** Lengthen the backtest and reserve a held-out period, selecting the holding horizon and hyperparameters on training data only, to confirm the edge is not an artifact of in-sample choices.
- **Richer analyst features.** Incorporate predictors identified in the literature, such as brokerage size, coverage breadth, and forecast boldness (6), to sharpen the strength estimate beyond the signal-to-drift relationship alone.
- **Impact-strength exploitation.** The current strategy trades only on drift strength. Combining impact and drift scores, or conditioning trade selection on both, may improve analyst selection.
- **Broader universe and regime testing.** Extend to small- and mid-cap names where drift may persist longer, and stress-test the strategy across distinct market regimes.

8. Conclusion

This project set out to evaluate the informational value of equity analyst recommendations on NASDAQ-100 constituents and to translate that value into a systematic trading strategy grounded in the Bayesian framework. Using the AnaChart dataset of analyst ratings and price targets, the work decomposed each recommendation into an immediate Impact component and

a delayed Drift component, estimated ex-ante Bayesian strength scores for individual analysts, and constructed a long-short strategy that trades the recommendations of underestimated analysts whose signals the market incorporates only gradually.

The empirical results support the central thesis. Post-recommendation drift remains exploitable in the NASDAQ-100 universe, and a strategy targeting it produced a positive, risk-adjusted return with a Sharpe ratio of 1.02 and an annualized return of 20.6% under signal-magnitude sizing, improving on the equal-weight baseline across every metric. Critically, the strategy achieved this with a near-zero market beta, delivering risk-adjusted performance comparable to the NASDAQ-100 benchmark while carrying materially lower volatility and drawdown, evidence that the returns stem from genuine analyst-signal alpha rather than inherited market exposure.

The analysis also refined the underlying thesis. The most heavily-covered analysts exhibit high impact strength and negligible drift, consistent with their information being immediately arbitrated away, while exploitable drift concentrates among less-covered analysts, precisely where the Bayesian shrinkage provides the most value as a safeguard against overfitting. Signal-magnitude sizing further demonstrated that the type of recommendation, a decisive rating change versus a marginal price-target revision, carries information beyond direction alone.

These findings are tempered by the limitations of a relatively short, single-regime backtest and a simplified market adjustment, and the borderline statistical significance of the estimated alpha counsels caution about the precise magnitude of the edge. Nonetheless, the project delivers a reproducible, interpretable, and look-ahead-safe methodology for extracting tradeable signals from analyst recommendations, and establishes that the selective targeting of underestimated analysts is a viable systematic approach in a universe where naive upgrade-downgrade strategies have long ceased to pay.

9. REFERENCES

- (1) Jaisson, T., Nguyen, L., Messikh, R. J., & Susinno, G. (2021). The Predictive Power of Analysts and Their Impact on Prices. SSRN Electronic Journal. <https://doi.org/10.2139/ssrn.3973080>
- (2) Barber, B. M., Lehavy, R., & Trueman, B. (2010). Ratings Changes, Ratings Levels, and the Predictive Value of Analysts' Recommendations. *Financial Management*, 39(2), 533–553. JSTOR. <https://doi.org/10.2307/40732448>
- (3) Vukovic, D. B., Ugolnikov, V., & Maiti, M. (2020). Sell-side analysts' recommendations a value or noise. *International Journal of Finance & Economics*, 26(2), 3134–3151. <https://doi.org/10.1002/ijfe.1954>
- (4) Lee, Y.-I., Hsieh, W.-L., & Miao, D. W.-C. (2024). A multi-dimensional assessment of the accuracy of analyst target prices. *International Review of Economics & Finance*, 93, 947–969. <https://doi.org/10.1016/j.iref.2024.02.056>
- (5) Vukovic, D. B., Orifjon O. U. Kurbonov, Moinak Maiti, Mustafa Özer, & Radovanovic, M. (2024). Outperforming the market: a comparison of Star and NonStar analysts' investment strategies and recommendations. *Humanities and Social Sciences Communications*, 11(1). <https://doi.org/10.1057/s41599-023-02527-8>
- (6) Jiang, S., Guo, Y., Zhou, W., & Li, X. (2022). Identifying predictors of analyst rating quality: An ensemble feature selection approach. *International Journal of Forecasting*, 39(4), 1853–1873. <https://doi.org/10.1016/j.ijforecast.2022.09.003>

- (7) Zaremba, A., & Konieczka, P. (2015). The profitability of following analyst recommendations on the Polish stock market. *RePEc: Research Papers in Economics*, 11(1), 22–31. https://doi.org/10.14636/1734-039x_11_1_003